

K L Deemed to be University Department of computer science and Engineering -1 -- KLBCH Course Handout 2026-2027, Odd Sem	
Course Title	: MACHINE LEARNING
Course Code	: 25SC2107E
L-T-P-S Structure	: 2-0-2-4
Pre-requisite	: 25SC1306E-COMPUTATIONAL FOUNDATIONS FOR ARTIFICIAL INTELLIGENCE
Credits	: 4
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Teaching Associates	:
Syllabus :	M1.PlacementPredict as a System — The ML Lifecycle A working prediction service running on day one — students issue requests a → trained model → packaged artifact → served endpoint → monitored production → retraining loop. • Where each lifecycle stage ac supervised vs unsupervised vs reinforcement learning split; what PlacementPredict is (supervised classification) and what it is not. • T boundary: the split that causes the most production bugs (training-serving skew); why feature stores exist. • A reading of the Placemec classical ML and ML engineering here; deep learning in DLAC; generative models in GALE; LLMs in NLPL. M2.Supervised Learn Regularisation: ridge (L2), lasso (L1), elastic net; the geometric picture of why L1 induces sparsity. • Logistic regression for binary cl problems. • Feature scaling and standardisation: when it matters (gradient descent), when it doesn't (tree models in M3); the Standard missing values: deletion, mean/median imputation, model-based imputation, MissingIndicator features. • A worked example end-to-e Tree-Based Models • Decision trees: greedy top-down construction, splitting criteria (Gini impurity, entropy, MSE for regression), stc subsampling, out-of-bag error, the variance-reduction argument. • Boosting: AdaBoost at intuition level, gradient boosting as functor importance: gini importance, permutation importance, SHAP values at preview level. • Tree models on tabular data: why they routine compare against linear baseline; interpret feature importance. M4.Unsupervised Learning • K-means revisited from CFAI with the mc datasets. • Hierarchical clustering: agglomerative bottom-up, the linkage choices (single, complete, average, Ward), reading dendrogr reduction revisited: PCA from CFAI in production-style usage; pitfalls when applied without standardisation. • t-SNE and UMAP as t SVM at intuition level, autoencoder-based anomaly detection at preview level (deep dive in DLAC). • Practical pipeline: when to clu Train/validation/test split discipline: the three-way split, why a test set must be held out from every modelling decision. • k-fold and s it lies), precision/recall/F1, the confusion matrix, ROC-AUC, PR-AUC; class-imbalance impact. • Regression metrics: RMSE, MAE, scaling, isotonic regression; reliability diagrams. • Hyperparameter search: grid search, random search (Bergstra-Bengio), Bayesian o plot. • Statistical significance for model comparison at intro level: McNemar's test for classifiers, paired bootstrap for regressors. M6. rituals that close the gap. • Feature engineering at scale: feature stores (Feast at conceptual level), the offline-online split, point-in-tim portable format, container-based serving. • Serving frameworks at intro level: FastAPI for REST, gRPC for low-latency, batch vs onli retraining, drift-triggered retraining, online learning at preview level. • A glance at MLOps platforms: MLflow for experiment trackin
Text Books :	Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow Aurélien Géron O'Reilly 2022 (3rd Ed.) The Elements of Sta An Introduction to Statistical Learning — with Applications in Python Gareth James, Daniela Witten, Trevor Hastie, Robert Tibshirar
Reference Books :	Pattern Recognition and Machine Learning Christopher M. Bishop Springer 2006 Christopher M. Bishop Springer 2006 Designing M Applications Emmanuel Ameisen O'Reilly 2020 Machine Learning Engineering Andriy Burkov True Positive Inc. 2020
MOOCS :	https://www.coursera.org/specializations/machine-learning-introduction https://www.coursera.org/specializations/mathematics-for-machine-learning https://www.coursera.org/specializations/mlops-machine-learning-duke?utm_medium=sem&utm_source=gg&utm_campaign=21104989118&adgroupid=&device=c&keyword=&matchtype=&network=x&device_model=&creativeid=&tp14bJaba&gclid=CjwKCAjwzevPBhBaEiwAplAxvmutse94FN3Buln0_4DOMCiBqBLflnwNErA8JvwpSXeh3zvLq4h_gxoCWQ
Course Rationale :	Machine Learning is the first course in the curriculum where students graduate from understanding intelligent systems conceptually to supervised classification system that predicts student placement outcomes from anonymised academic and co-curricular data. Student orientation is deliberate: it mirrors the way ML is practised in industry, where engineers inherit a running system and progressively ur
Course Objectives :	1. Machine Learning is the first course where students build, evaluate, and deploy real ML systems. The PBL track is OUTSIDE-IN – prediction service (PlacementPredict, predicting student placement outcomes from anonymised academic data) — issuing requests, g belong to GALE (T05); NLP belongs to NLPL (T08). MLNG owns the classical-ML canon (linear models, tree ensembles, clustering training-serving skew, drift, basic MLOps). Pre-requisite: CFAI, DSA-2, PSPJ. Enables: GALE, DLAC, NLPL, AAAE — and placen Meta, Adobe), Indian unicorns (Razorpay, Swiggy, Zomato, CRED), and AI-first startups.
Global Certifications :	1. NVIDIA DLI Computer Vision Specialization - NVIDIA Deep Learning Institute 2. AWS ML Engineer – Associate (MLA-C01) -

COURSE OUTCOMES (COs):

CO NO	Course Outcome (CO)	PO/PSO	Blooms Taxonomy Level (BTL)
CO1	Analyze the end-to-end ML system lifecycle — data → features → training → evaluation → deployment → monitoring — and trace one prediction request from a deployed model back through every preceding lifecycle stage.	PO2,PSO1,PO3,PO1	4
CO2	Apply linear supervised-learning models — linear regression, ridge, lasso, elastic net, logistic regression, multinomial logistic regression — and reason about regularisation, the bias-variance trade-off, and feature scaling.	PO1,PO2,PO3,PSO1	3
CO3	Apply tree-based supervised-learning models — decision trees, random forests, gradient-boosted trees (XGBoost, LightGBM) — and reason about feature importance, splitting criteria, and ensemble variance reduction.	PO1,PO3,PSO1	3
CO4	Apply unsupervised-learning techniques — k-means and hierarchical clustering, density-based clustering (DBSCAN), dimensionality reduction (PCA, t-SNE, UMAP), and anomaly detection — to discover structure in unlabelled data.	PO1,PSO1,PO2	3
CO5	Analyze model performance using cross-validation, train/validation/test splits, classification metrics (accuracy, precision, recall, F1, ROC-AUC, PR-AUC), regression metrics (RMSE, MAE, R ²), calibration, and hyperparameter search (grid, random, Bayesian).	PO1,PO2,PSO1,PO4	4
CO6	Apply ML-engineering practices — feature stores, training-serving skew avoidance, model packaging, REST-based serving, basic monitoring for drift and performance degradation — to deploy a production-style ML system.	PO1,PO3,PSO1,PO5	3

COURSE OUTCOME INDICATORS (COIs):

Outcome No.	Highest BTL	COI-2	COI-3	COI-4
CO1	4	Btl-2 Describe each stage of the ML lifecycle and identify which PlacementPredict component corresponds to each stage.	Btl-3 Execute a trace of one HTTP prediction request through the PlacementPredict system, identifying the role of the feature pipeline, inference engine, and response formatter.	Btl-4 Examine the training-serving boundary in PlacementPredict, diagnose a simulated skew scenario, and propose a feature store strategy to eliminate it.
CO2	3	Btl-2 Explain the geometric interpretation of L1 and L2 regularisation, the role of the sigmoid function in logistic regression, and the effect of feature scaling on gradient descent convergence.	Btl-3 Implement ridge, lasso, and logistic regression on the PlacementPredict dataset; tune regularisation hyperparameters using cross-validation; and interpret learned coefficients.	
CO3	3	Btl-2 Explain Gini impurity and entropy as splitting criteria, describe the bagging and feature-subsampling mechanism in random forests, and outline how gradient boosting frames each successive tree as a functional gradient step.	Btl-3 Train a decision tree, a random forest, and an XGBoost model on the placement dataset; compare OOB error against held-out validation loss; and extract permutation importance and SHAP values.	
CO4	3	Btl-2 Describe the k-means objective function, elbow method, silhouette score, density-reachability logic of DBSCAN, and the geometric intuition of PCA variance maximisation.	Btl-3 Apply k-means, agglomerative hierarchical clustering, and DBSCAN to the student feature matrix; generate a dendrogram; reduce dimensionality with PCA; and visualise cluster separation using t-SNE.	
CO5	4	Btl-2 Explain why a separate test set must be held out from all modelling decisions, when ROC-AUC can be misleading for imbalanced classes, and what a calibrated probability output means for downstream decision-making.	Btl-3 Implement stratified k-fold cross-validation on PlacementPredict; compute precision, recall, F1, ROC-AUC, and PR-AUC; produce a calibration reliability diagram; and run random search hyperparameter optimisation.	Btl-4 Analyze learning curves to diagnose underfitting and overfitting in two competing models, interpret the PR-AUC advantage under class imbalance, and apply McNemar's test to determine whether the performance difference is statistically significant.
CO6	3	Btl-2 Describe the offline-online feature store split, explain training-serving skew as the most common production ML bug class, and identify the role of MLflow experiment tracking and Evidently AI drift monitoring.	Btl-3 Package the trained PlacementPredict model using ONNX; serve it via a FastAPI REST endpoint; log experiments to MLflow; and implement a basic data drift monitor using Evidently AI.	

PROGRAM OUTCOMES & PROGRAM SPECIFIC OUTCOMES (POs/PSOs)

Po No.	Program Outcome
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Lecture Course DELIVERY Plan:

Sess.No.	CO	COI	Topic	Book No[CH No][Page No]	Teaching-Learning Methods	EvaluationComponents
1	CO1	COI-2	Introduction to PlacementPredict – live prediction service, machine learning categories, supervised binary classification	TB1: Ch.1, pp. 3–37	PPT,Talk	AI-Enabled Adaptive Learning Engagement,End Semester Exam,Global Challenges - Participation,Mid-Term Examination (Descriptive),MOOCs Progress Review
2	CO1	COI-2	Machine Learning System Lifecycle – data to deployment pipeline and codebase mapping	TB1: Ch.1, pp. 3–37 RB2: Ch.1, pp. 1–20	PPT,Talk	AI-Enabled Adaptive Learning Engagement,End Semester Exam,Global Challenges - Participation,Mid-Term Examination (Descriptive)
3	CO1	COI-3	End-to-End Prediction Flow – request processing, inference pipeline, batch vs real-time prediction, training-serving boundary	TB1: Ch.1, pp. 3–37 RB2: Ch.2, pp. 21–58	PPT,Talk	AI-Enabled Adaptive Learning Engagement,End Semester Exam,Global Challenges - Participation,Mid-Term Examination (Descriptive)
4	CO1	COI-4	PlacementPredict Codebase Tour – ingestion, feature engineering, training, serving, monitoring and course scope	RB2: Ch.5, pp. 133–172 RB4: Ch.3, pp. 45–78	PPT,Talk	AI-Enabled Adaptive Learning Engagement,End Semester Exam,Global Challenges - Participation,Mid-Term Examination (Descriptive)
5	CO2	COI-2	Linear Regression – MSE cost function, normal equations, gradient descent training, geometry of fitted hyperplane	TB1: Ch.4, pp. 107–136 TB2: Ch.2, pp. 9–16; Ch.3, pp. 43–99	PPT,Talk	AI-Enabled Adaptive Learning Engagement,End Semester Exam,Global Challenges - Participation,Mid-Term Examination

Sess.No.	CO	COI	Topic	Book No[CH No][Page No]	Teaching-Learning Methods	EvaluationComponents
						(Descriptive),MOOCs Progress Review
6	CO2	COI-3	Logistic Regression – sigmoid function, cross-entropy loss, decision boundaries, multinomial logistic regression (softmax)	TB1: Ch.4, pp. 148–166 TB3: Ch.4, pp. 129–174	PPT,Talk	AI-Enabled Adaptive Learning Engagement,End Semester Exam,Global Challenges - Participation,Mid-Term Examination (Descriptive)
7	CO2	COI-2	Regularisation – Ridge, Lasso, Elastic Net, sparsity, bias-variance tradeoff	TB1: Ch.4, pp. 136–147 TB2: Ch.3, pp. 57–79 TB3: Ch.6, pp. 225–264	PPT,Talk	AI-Enabled Adaptive Learning Engagement,End Semester Exam,Global Challenges - Participation,Mid-Term Examination (Descriptive)
8	CO2	COI-3	Feature Engineering – scaling, encoding techniques, embedding preview, missing value imputation methods	TB1: Ch.2, pp. 67–89 TB3: Ch.3, pp. 71–120	LTC,PPT	AI-Enabled Adaptive Learning Engagement,End Semester Exam,Global Challenges - Participation,Mid-Term Examination (Descriptive)
9	CO3	COI-2	Decision Trees – tree construction, splitting criteria, pruning techniques, bias-variance considerations	TB1: Ch.6, pp. 185–214 TB3: Ch.8, pp. 303–336	PPT,Talk	AI-Enabled Adaptive Learning Engagement,End Semester Exam,Global Challenges - Participation,Mid-Term Examination (Descriptive),MOOCs Progress Review
10	CO3	COI-3	Random Forests – bagging, feature subsampling, out-of-bag error, ensemble learning	TB1: Ch.7, pp. 215–247 TB3: Ch.8, pp. 336–355	LTC,PPT	AI-Enabled Adaptive Learning Engagement,End Semester Exam,Global Challenges - Participation,Mid-Term Examination (Descriptive)
11	CO3	COI-3	Boosting Techniques – AdaBoost, Gradient Boosting, XGBoost, LightGBM	TB1: Ch.7, pp. 247–264 TB2: Ch.10, pp. 337–387	LTC,PPT	AI-Enabled Adaptive Learning Engagement,End Semester Exam,Global Challenges - Participation,Mid-Term Examination (Descriptive)
12	CO3	COI-3	Feature Importance and Model Explainability – MDI, permutation importance, SHAP preview, tree models vs neural networks	TB1: Ch.7, pp. 240–247 RB2: Ch.5, pp. 133–172	PPT,Talk	AI-Enabled Adaptive Learning Engagement,End Semester Exam,Global Challenges - Participation,Mid-Term Examination (Descriptive)
13	CO4	COI-2	K-Means Clustering – Lloyd algorithm, K-Means++, mini-batch K-Means, cluster evaluation methods	TB1: Ch.9, pp. 295–316 TB3: Ch.12, pp. 497–530	PPT,Talk	AI-Enabled Adaptive Learning Engagement,End Semester Exam,Global Challenges - Participation,Mid-Term Examination (Descriptive),MOOCs Progress Review
14	CO4	COI-2	Hierarchical Clustering and DBSCAN – linkage methods, dendrogram analysis, density-based clustering	TB1: Ch.9, pp. 316–328 TB2: Ch.14, pp. 501–528	PPT,Talk	AI-Enabled Adaptive Learning Engagement,End Semester Exam,Global Challenges - Participation,Mid-Term Examination (Descriptive)
15	CO4	COI-3	Dimensionality Reduction – PCA, t-SNE, UMAP, visualization and feature compression	TB1: Ch.8, pp. 265–294 TB2: Ch.14, pp. 534–570	PPT,Talk	AI-Enabled Adaptive Learning Engagement,End Semester Exam,Global Challenges - Participation,Mid-Term Examination (Descriptive)
16	CO4	COI-3	Anomaly Detection and Advanced Feature Engineering –cluster features	TB1: Ch.9, pp. 295–316 TB3: Ch.12, pp. 497–530	PPT,Talk	AI-Enabled Adaptive Learning Engagement,End Semester Exam,Global Challenges - Participation,Mid-Term Examination (Descriptive)
17	CO5	COI-2	Model Evaluation Strategy – train/validation/test split, data leakage prevention, cross-validation techniques	TB1: Ch.2, pp. 85–106 TB3: Ch.5, pp. 197–224	PPT,Talk	AI-Enabled Adaptive Learning Engagement,End Semester Exam,Global Challenges - Participation,Mid-Term Examination (Descriptive),MOOCs Progress Review

Sess.No.	CO	COI	Topic	Book No[CH No][Page No]	Teaching-Learning Methods	EvaluationComponents
18	CO5	COI-3	Performance Metrics – classification metrics, ROC-AUC, PR-AUC, regression metrics and model assessment	TB1: Ch.3, pp. 83–106 TB3: Ch.4, pp. 174–196	PPT,Talk	AI-Enabled Adaptive Learning Engagement,End Semester Exam,Global Challenges - Participation,Mid-Term Examination (Descriptive)
19	CO5	COI-4	Hyperparameter Optimization – Grid Search, Random Search, Bayesian Optimization, learning and validation curves	TB1: Ch.2, pp. 67–89; Ch.3, pp. 83–106 TB2: Ch.7, pp. 219–258	PPT,Talk	AI-Enabled Adaptive Learning Engagement,End Semester Exam,Global Challenges - Participation,Mid-Term Examination (Descriptive)
20	CO5	COI-4	Probability Calibration and Statistical Validation – calibration methods, reliability diagrams, significance testing	TB1: Ch.2, pp. 67–89; Ch.3, pp. 83–106 TB2: Ch.7, pp. 219–258	PPT,Talk	AI-Enabled Adaptive Learning Engagement,End Semester Exam,Global Challenges - Participation,Mid-Term Examination (Descriptive)
21	CO6	COI-2	ML Engineering Foundations – notebook-to-production transition, modular pipelines, configuration management, testing and logging	RB2: Ch.6, pp. 173–210; Ch.7, pp. 211–246 RB4: Ch.5, pp. 103–138	PPT,Talk	AI-Enabled Adaptive Learning Engagement,End Semester Exam,Global Challenges - Participation,Mid-Term Examination (Descriptive),MOOCs Progress Review
22	CO6	COI-2	Feature Stores and Training-Serving Consistency – Feast concepts, point-in-time correctness, training-serving skew prevention	RB2: Ch.6, pp. 173–210; Ch.7, pp. 211–246 RB4: Ch.5, pp. 103–138	PPT,Talk	AI-Enabled Adaptive Learning Engagement,End Semester Exam,Global Challenges - Participation,Mid-Term Examination (Descriptive)
23	CO6	COI-3	Model Packaging and Deployment – Joblib, ONNX, FastAPI, REST vs gRPC, batch and online prediction systems	RB2: Ch.8, pp. 247–286; Ch.9, pp. 287–318 RB4: Ch.7, pp. 163–196; Ch.8, pp. 197–228	LTC,PPT	AI-Enabled Adaptive Learning Engagement,End Semester Exam,Global Challenges - Participation,Mid-Term Examination (Descriptive)
24	CO6	COI-3	Monitoring and Lifecycle Management – drift detection, retraining strategies, MLflow, DVC and production observability	RB2: Ch.8, pp. 247–286; Ch.9, pp. 287–318 RB4: Ch.7, pp. 163–196; Ch.8, pp. 197–228	LTC,PPT	AI-Enabled Adaptive Learning Engagement,End Semester Exam,Global Challenges - Participation,Mid-Term Examination (Descriptive)
25	CO6	COI-3	Revision	RB2: Ch.8, pp. 247–286; Ch.9, pp. 287–318 RB4: Ch.7, pp. 163–196; Ch.8, pp. 197–228	PPT,Talk	AI-Enabled Adaptive Learning Engagement,End Semester Exam,Global Challenges - Participation,Mid-Term Examination (Descriptive)

Lecture Session wise Teaching – Learning Plan

SESSION NUMBER : 1

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching- Learning Methods	Active Learning Methods
5	Attendance and Course Orientation	1	Talk	--- NOT APPLICABLE ---
20	Interact with the PlacementPredict service by submitting sample student profiles and observing prediction outputs. Based on the inputs and responses, they discuss how the system makes decisions and identify the type of problem it appears to solve	2	PPT	--- NOT APPLICABLE ---
20	Introduction to supervised, unsupervised, and reinforcement learning using practical examples. PlacementPredict is analyzed as a supervised binary classification system, and students distinguish it from clustering, recommendation, and generative AI applications.	3	PPT	--- NOT APPLICABLE ---
5	Conclusion and Summary	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 2

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching- Learning Methods	Active Learning Methods
5	Attendance and Recap	1	Talk	--- NOT APPLICABLE ---

Time(min)	Topic	BTL	Teaching- Learning Methods	Active Learning Methods
20	Examine how PlacementPredict transforms raw student records into predictions through data collection, preprocessing, feature engineering, model training, deployment, monitoring, and retraining activities.	2	PPT	--- NOT APPLICABLE ---
20	Explanation of the machine learning lifecycle and the role of each stage. Students identify where datasets, features, trained models, deployment artifacts, and monitoring components are located within the PlacementPredict project structure.	2	PPT	--- NOT APPLICABLE ---
5	Conclusion and Summary	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 3

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching- Learning Methods	Active Learning Methods
5	Attendance and Recap	1	Talk	--- NOT APPLICABLE ---
20	Trace a prediction request through PlacementPredict from API submission to prediction response. They observe how user inputs are validated, transformed into features, passed to the model, and returned as prediction results.	3	PPT	--- NOT APPLICABLE ---
20	Discussion of real-time and batch prediction workflows using placement prediction examples. Students examine the training-serving boundary and understand how inconsistencies between training and deployment environments can introduce prediction errors in production systems.	3	PPT	--- NOT APPLICABLE ---
5	Conclusion and Summary	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 4

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching- Learning Methods	Active Learning Methods
5	Attendance and Recap	1	Talk	--- NOT APPLICABLE ---
20	Explore the PlacementPredict project structure and identify the responsibilities of the data ingestion module, feature engineering pipeline, model training script, serving application, and monitoring components.	4	PPT	--- NOT APPLICABLE ---
20	Discussion of the scope of this course and its relationship with Deep Learning, Generative AI, and Large Language Models. Students identify where traditional machine learning ends and where advanced AI courses begin.	4	PPT	Brain Storming
5	Conclusion and Summary	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 5

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching- Learning Methods	Active Learning Methods
5	Attendance and Recap	1	Talk	--- NOT APPLICABLE ---
20	Analyze how PlacementPredict could estimate a continuous outcome such as expected salary package or placement score. They examine relationships between input variables and target values and observe how a best-fit prediction model can be constructed from historical placement data.	2	PPT	--- NOT APPLICABLE ---
20	Introduction to linear regression, hypothesis function, Mean Squared Error (MSE), normal equations, and gradient descent optimization. Students visualize the fitted hyperplane and understand how model parameters are adjusted to minimize prediction error.	2	PPT	--- NOT APPLICABLE ---
5	Conclusion and Summary	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 6

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching- Learning Methods	Active Learning Methods
5	Attendance and Recap	1	Talk	--- NOT APPLICABLE ---
20	Revisit PlacementPredict and examine how the system predicts whether a student is likely to be placed or not placed. They compare regression-style outputs with classification decisions and discuss the need for probability-based predictions.	3	PPT	Immediate feedback
20	Introduction to logistic regression, sigmoid function, probability estimation, cross-entropy loss, and linear decision boundaries. Students explore multinomial logistic regression using softmax and understand how multi-class classification problems are handled.	3	PPT	--- NOT APPLICABLE ---
5	Conclusion and Summary	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 7

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching- Learning Methods	Active Learning Methods
5	Attendance and Recap	1	Talk	--- NOT APPLICABLE ---
20	Analyze a PlacementPredict model trained with a large number of features and observe situations where the model performs well on training data but poorly on unseen data. The need to control model complexity is discussed through practical examples.	2	PPT	Leading question
20	Introduction to Ridge (L2), Lasso (L1), and Elastic Net regularisation. Students examine the geometric intuition behind sparsity, compare feature selection behavior, and understand the bias-variance tradeoff in predictive modeling.	3	PPT	--- NOT APPLICABLE ---
5	Conclusion and Summary	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 8

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching- Learning Methods	Active Learning Methods
5	Attendance and Recap	1	Talk	--- NOT APPLICABLE ---
20	Inspect PlacementPredict input data and identify numerical, categorical, ordinal, and missing-value attributes. They discuss why raw data cannot be directly consumed by machine learning algorithms and explore preprocessing requirements.	3	PPT	--- NOT APPLICABLE ---
20	Introduction to StandardScaler, MinMaxScaler, one-hot encoding, ordinal encoding, target encoding, embedding-based encoding concepts, and missing value imputation techniques including mean, median, model-based methods, and MissingIndicator features.	3	PPT	--- NOT APPLICABLE ---
5	Conclusion and Summary	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 9

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching- Learning Methods	Active Learning Methods
5	Attendance and Recap	1	Talk	--- NOT APPLICABLE ---
20	Analyze historical PlacementPredict data and observe how placement decisions can be represented as a sequence of questions based on academic performance, skills, internships, and project experience. They identify how repeated data partitioning leads to interpretable decision-making paths.	2	PPT	--- NOT APPLICABLE ---
20	Introduction to greedy top-down tree construction, Gini impurity, entropy, and MSE-based splitting. Students examine stopping criteria, pre-pruning and post-pruning techniques, and understand the bias-variance tradeoff associated with shallow and deep decision trees.	3	PPT	--- NOT APPLICABLE ---
5	Conclusion and Summary	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 10

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching- Learning Methods	Active Learning Methods
5	Attendance and Recap	1	Talk	--- NOT APPLICABLE ---
20	Compare predictions from a single PlacementPredict decision tree with predictions obtained from multiple trees trained on different subsets of data. They observe variations in decision paths and discuss how combining models improves reliability.	2	PPT	--- NOT APPLICABLE ---
20	Introduction to bagging, bootstrap sampling, feature subsampling, and out-of-bag error estimation. Students explore how random forests reduce variance, improve generalization, and outperform individual deep trees in many tabular-data applications.	3	PPT	--- NOT APPLICABLE ---
5	Conclusion and Summary	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 11

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching- Learning Methods	Active Learning Methods
5	Attendance and Recap	1	Talk	--- NOT APPLICABLE ---
20	Examine PlacementPredict cases that are repeatedly misclassified by a model. They investigate how additional models can focus on difficult examples and gradually improve overall prediction accuracy through iterative learning.	3	PPT	--- NOT APPLICABLE ---
20	Introduction to AdaBoost, Gradient Boosting, XGBoost, and LightGBM. Students learn sample reweighting, functional gradient descent intuition, histogram-based splitting, regularisation, leaf-wise growth strategies, and performance advantages of modern boosting frameworks.	3	PPT	--- NOT APPLICABLE ---
5	Conclusion and Summary 4 Talk --- NOT APPLICABLE --- SESSION NUMBER : 12 Session Outcome: Students will understand feature importance techniques and interpret model predictions using explainability methods for tabular machine learning systems. Time(min)	1	Talk	--- NOT APPLICABLE ---

Time(min)	Topic	BTL	Teaching- Learning Methods	Active Learning Methods
	Topic BTL Teaching-Learning Methods Active Learning Methods 5 Attendance and Recap 1 Talk --- NOT APPLICABLE --- 20 Students analyze a trained PlacementPredict model and identify which features contribute most to placement decisions. They compare the influence of academic, technical, and experiential attributes and discuss how feature importance supports model interpretation. 4 PPT, Demonstration One Minute Paper 20 Introduction to Gini-based importance, permutation importance, and SHAP value interpretation. Students explore SHAP beeswarm visualizations and compare tree-based models with neural networks for structured tabular data, reflecting common practices in real-world machine learning competitions and industry projects. 4 PPT, Talk Guided Discussion 5 Conclusion and Summary 4 Talk --- NOT APPLICABLE --- Module 3 Learning Flow Session 9: How a single tree learns decision rules → Session 10: How multiple trees combine to improve performance → Session 11: How boosting sequentially corrects errors → Session 12: How model decisions are explained and interpreted This progression mirrors the evolution of a real PlacementPredict system—from a simple interpretable model to powerful ensemble methods and finally to explainable AI techniques used in production ML systems.			

SESSION NUMBER : 12

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching- Learning Methods	Active Learning Methods
5	Attendance and Recap	1	Talk	--- NOT APPLICABLE ---
20	Analyze a trained PlacementPredict model and identify which features contribute most to placement decisions. They compare the influence of academic, technical, and experiential attributes and discuss how feature importance supports model interpretation.	2	PPT	--- NOT APPLICABLE ---
20	Introduction to Gini-based importance, permutation importance, and SHAP value interpretation. Students explore SHAP beeswarm visualizations and compare tree-based models with neural networks for structured tabular data, reflecting common practices in real-world machine learning competitions and industry projects.	3	PPT	--- NOT APPLICABLE ---
5	Conclusion and Summary	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 13

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching- Learning Methods	Active Learning Methods
5	Attendance and Recap	1	Talk	--- NOT APPLICABLE ---
20	Analyze PlacementPredict data without placement labels and attempt to identify natural groupings of students based on academic performance, skills, projects, and internship profiles. They explore how similar students can be grouped into clusters for pattern discovery and segmentation.	2	PPT	--- NOT APPLICABLE ---
20	Introduction to K-Means clustering, Lloyd's algorithm, K-Means++ initialization, and mini-batch K-Means. Students examine cluster quality using the elbow method and silhouette score and understand the impact of selecting different values of K.	3	PPT	--- NOT APPLICABLE ---
5	Conclusion and Summary	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 14

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching- Learning Methods	Active Learning Methods
5	Attendance and Recap	1	Talk	--- NOT APPLICABLE ---
20	Investigate groups of students with overlapping characteristics and observe situations where cluster shapes are irregular. They compare clustering outcomes and discuss how different grouping strategies reveal hidden patterns within PlacementPredict data.	3	PPT	--- NOT APPLICABLE ---
20	Introduction to agglomerative hierarchical clustering, linkage strategies, dendrogram interpretation, and cluster extraction. Students then explore DBSCAN concepts including ϵ , minPts, core points, border points, noise points, and clustering of non-convex data structures.	3	PPT	Case Study Based Learning
5	Conclusion and Summary	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 15

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching- Learning Methods	Active Learning Methods
5	Attendance and Recap	1	Talk	--- NOT APPLICABLE ---
20	Examine PlacementPredict datasets containing multiple features and discuss challenges associated with high-dimensional data. They explore how dimensionality reduction can simplify visualization and reveal hidden structures while retaining important information.	2	PPT	--- NOT APPLICABLE ---
20	Introduction to PCA, explained variance ratio, scree plots, and the importance of feature standardization. Students compare PCA with t-SNE and UMAP, exploring key parameters	3	PPT	--- NOT APPLICABLE ---

Time(min)	Topic	BTL	Teaching- Learning Methods	Active Learning Methods
	such as perplexity, n_neighbors, and min_dist while understanding their strengths and limitations for data visualization.			
5	Conclusion and Summary	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 16

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching- Learning Methods	Active Learning Methods
5	Attendance and Recap	1	Talk	--- NOT APPLICABLE ---
20	Analyze PlacementPredict records containing unusual academic or skill profiles and identify observations that significantly differ from normal patterns. They discuss the impact of such records on model training, prediction quality, and decision-making reliability.	3	PPT	--- NOT APPLICABLE ---
20	Explore how cluster assignments generated through unsupervised learning can be incorporated as engineered features in downstream supervised prediction models.	3	PPT	Debates
5	Conclusion and Summary	1	PPT	--- NOT APPLICABLE ---

SESSION NUMBER : 17

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching- Learning Methods	Active Learning Methods
5	Attendance and Recap	1	Talk	--- NOT APPLICABLE ---
20	Evaluate PlacementPredict models using different dataset partitions and observe how model performance changes when training, validation, and testing data are not properly separated. They discuss practical examples of data leakage and its impact on reported accuracy.	2	PPT	--- NOT APPLICABLE ---
20	Introduction to train, validation, and test splits, data leakage prevention, k-fold cross-validation, stratified k-fold validation, and nested cross-validation. Students understand why the test set must remain isolated from all model-selection decisions.	3	PPT	--- NOT APPLICABLE ---
5	Conclusion and Summary	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 18

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching- Learning Methods	Active Learning Methods
5	Attendance and Recap	1	Talk	--- NOT APPLICABLE ---
20	Analyze PlacementPredict predictions and compare model performance under balanced and imbalanced class distributions. They investigate prediction outcomes using confusion matrices and identify situations where accuracy may provide misleading conclusions.	4	PPT	--- NOT APPLICABLE ---
20	Introduction to accuracy, precision, recall, F1-score, confusion matrix, ROC-AUC, and PR-AUC. Students also examine RMSE, MAE, MAPE, and R ² for regression tasks and compare the sensitivity of RMSE and MAE to outliers.	3	PPT	--- NOT APPLICABLE ---
5	Conclusion and Summary	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 19

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching- Learning Methods	Active Learning Methods
5	Attendance and Recap	1	Talk	--- NOT APPLICABLE ---
20	Compare multiple PlacementPredict model configurations and observe how parameter choices affect predictive performance. They analyze examples of underfitting and overfitting and discuss the challenges of selecting optimal model settings manually.	3	PPT	--- NOT APPLICABLE ---
20	Introduction to GridSearchCV, RandomizedSearchCV, and Bayesian optimization concepts. Students interpret learning curves and validation curves to identify underfitting, overfitting, and data scarcity issues in machine learning systems.	4	PPT	--- NOT APPLICABLE ---
5	Conclusion and Summary	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 20

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching- Learning Methods	Active Learning Methods
5	Attendance and Recap	1	Talk	--- NOT APPLICABLE ---
20	Examine PlacementPredict probability outputs and compare prediction confidence with actual outcomes. They discuss whether a predicted probability truly reflects real-world likelihood	3	PPT	--- NOT APPLICABLE ---

Time(min)	Topic	BTL	Teaching- Learning Methods	Active Learning Methods
	and investigate situations where models appear accurate but poorly calibrated.			
20	Introduction to probability calibration using Platt scaling and isotonic regression. Students interpret reliability diagrams and explore statistical significance testing using McNemar's test for classification models and paired bootstrap methods for regression models.	4	PPT	One minute paper
5	Conclusion and Summary	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 21

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching- Learning Methods	Active Learning Methods
5	Attendance and Recap	1	Talk	--- NOT APPLICABLE ---
20	Compare a PlacementPredict model developed in a Jupyter notebook with the requirements of a production-ready system. They identify challenges related to reproducibility, maintainability, configuration management, testing, and deployment when moving beyond experimentation environments.	2	PPT	--- NOT APPLICABLE ---
20	Modular ML pipelines, script-based workflows, argparse and YAML configuration management, structured logging, and unit testing for machine learning components. Students understand how engineering practices improve reliability, scalability, and maintainability of ML systems.	3	PPT	--- NOT APPLICABLE ---
5	Conclusion and Summary	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 22

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching- Learning Methods	Active Learning Methods
5	Attendance and Recap	1	Talk	--- NOT APPLICABLE ---
20	Investigate scenarios where PlacementPredict performs well during training but produces inaccurate predictions after deployment. They analyze how differences in feature generation and data availability create inconsistencies between training and serving environments.	2	PPT	--- NOT APPLICABLE ---
20	Introduction to feature stores, offline and online feature repositories, point-in-time correctness, and training-serving skew. Students explore how reusable preprocessing pipelines and shared ColumnTransformer objects help maintain consistency between training and production systems.	2	PPT	Fish Bowl
5	Conclusion and Summary	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 23

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching- Learning Methods	Active Learning Methods
5	Attendance and Recap	1	Talk	--- NOT APPLICABLE ---
20	Examine how a trained PlacementPredict model can be transformed into a reusable service capable of serving predictions to multiple users. They explore deployment requirements, portability concerns, and integration with software applications.	3	PPT	--- NOT APPLICABLE ---
20	Introduction to model serialization using joblib, limitations of pickle, ONNX for cross-platform deployment, FastAPI-based serving, REST and gRPC communication concepts, and architectural differences between batch prediction and online prediction systems.	3	PPT	--- NOT APPLICABLE ---
5	Conclusion and Summary	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 24

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching- Learning Methods	Active Learning Methods
5	Attendance and Recap	1	Talk	--- NOT APPLICABLE ---
20	Analyze the performance of a deployed PlacementPredict service over time and investigate situations where prediction quality gradually degrades. They discuss operational challenges arising from changing data patterns, user behavior, and business requirements.	2	PPT	--- NOT APPLICABLE ---
20	Introduction to performance monitoring, latency, throughput, data drift, concept drift, label drift, and Population Stability Index (PSI). Students explore retraining strategies, experiment tracking with MLflow, and data versioning using DVC for managing production ML lifecycles.	3	PPT	Seminars
5	Conclusion and Summary	1	PPT	--- NOT APPLICABLE ---

SESSION NUMBER : 25

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching- Learning Methods	Active Learning Methods
5	Attendance and Recap	1	Talk	--- NOT APPLICABLE ---
20	MLflow tracking — log hyperparameters, validation AUC, calibration ECE, and the ONNX artifact to an MLflow run; register the best model in the model registry with the Production tag	2	PPT	--- NOT APPLICABLE ---
20	End-to-end demo — complete PlacementPredict serving pipeline runs live; students issue requests to their own deployed endpoint	3	PPT	--- NOT APPLICABLE ---
5	Conclusion and Summary	1	LTC	--- NOT APPLICABLE ---

Tutorial Course DELIVERY Plan:

NO Delivery Plan Exists

Tutorial Session wise Teaching – Learning Plan

No Session Plans Exists

Practical Course DELIVERY Plan:

Tutorial Session no	Topics	CO-Mapping
1	Environment Setup and Dataset Exploration	CO1
2	Experiment: Exploratory Data Analysis and Visualisation Report	CO1
3	Gradient Descent from Scratch — CGPA Regression	CO2
4	Logistic Regression — Binary and Multinomial Classification	CO2
5	Ridge, Lasso, and Elastic Net Regularisation	CO2
6	Decision Trees — Splitting Criteria, CCP Pruning, and Depth Sweep	CO3
7	Experiment: Random Forests — OOB Error, Bagging, and Feature Subsampling	CO3
8	XGBoost, LightGBM, and SHAP — PlacementPredict	CO3
9	Clustering — K-Means, Hierarchical, and DBSCAN	CO4
10	Dimensionality Reduction and Anomaly Detection	CO4
11	Data Leakage, Cross-Validation, and Nested CV	CO5
12	Full Metrics Suite — Confusion Matrix, ROC-AUC, PR-AUC, Calibration, McNemar	CO5
13	Modular Pipeline, FastAPI Endpoint, ONNX Conversion, and MLflow Tracking	CO6

Practical Session wise Teaching – Learning Plan

SESSION NUMBER : 1

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching- Learning Methods	Active Learning Methods
5	Attendance	1	Talk	--- NOT APPLICABLE ---
20	Load placement_predict_50k.csv; inspect shape, dtypes, describe(); class distribution of PlacementStatus; library version verification	3	PPT	--- NOT APPLICABLE ---
25	EXECUTION, Analysis and viva	4	PPT	--- NOT APPLICABLE ---
20	Write summarise_dataset(df) — prints shape, dtype groups, missing value count, class distribution for any DataFrame	3	PPT	--- NOT APPLICABLE ---
25	EXECUTION, Analysis and viva	4	PPT	--- NOT APPLICABLE ---
5	Conclusion and Summary	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 2

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching- Learning Methods	Active Learning Methods
5	Attendance and Recap	1	Talk	--- NOT APPLICABLE ---
20	Missing value heatmap; histograms; countplot by Stream; correlation heatmap; boxplots; multi-page PDF export using PdfPages.	3	PPT	--- NOT APPLICABLE ---
25	EXECUTION, Analysis and Viva	4	PPT	--- NOT APPLICABLE ---
20	Export 5-page EDA PDF using PdfPages: missingness heatmap, 6 histograms, placement rate by stream, correlation heatmap, boxplots.	3	PPT	--- NOT APPLICABLE ---
25	EXECUTION, Analysis and Viva	4	PPT	--- NOT APPLICABLE ---
5	Conclusion and Summary	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 3

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching- Learning Methods	Active Learning Methods
5	Attendance and Recap	1	Talk	--- NOT APPLICABLE ---
20	Batch gradient descent in NumPy; MSE loss curve; compare GD vs sklearn LinearRegression RMSE; mini-batch GD extension	3	PPT	--- NOT APPLICABLE ---
25	EXECUTION, Analysis and Viva	4	PPT	--- NOT APPLICABLE ---
20	Implement mini-batch GD (batch_size=32, 1000 epochs); plot alongside batch GD loss; assert mini-batch RMSE within 0.3 of sklearn	3	PPT	--- NOT APPLICABLE ---
25	EXECUTION, Analysis and Viva	4	PPT	--- NOT APPLICABLE ---
5	Conclusion and Summary	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 4

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching- Learning Methods	Active Learning Methods
5	Attendance and Recap	1	Talk	--- NOT APPLICABLE ---
20	OHE + scaling pipeline; binary PlacementStatus LR; multinomial CGPA_Tier LR; coefficient bar chart; evaluate_classifier() function	3	PPT	--- NOT APPLICABLE ---
25	EXECUTION, Analysis and Viva	4	PPT	--- NOT APPLICABLE ---
20	Write evaluate_classifier(model, X_train, y_train, X_val, y_val) — fits and prints train accuracy, val accuracy, classification_report for 3 models	3	PPT	--- NOT APPLICABLE ---
25	EXECUTION, Analysis and Viva	4	PPT	--- NOT APPLICABLE ---
5	Conclusion and Summary 4	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 5

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching- Learning Methods	Active Learning Methods
5	Attendance and Recap	1	Talk	--- NOT APPLICABLE ---
20	Lasso/Ridge/ElasticNet shrinkage paths; validation accuracy vs C; nonzero coefficient count; compare_regularisation() DataFrame.	3	PPT	--- NOT APPLICABLE ---
25	EXECUTION, Analysis and Viva	4	PPT	--- NOT APPLICABLE ---
20	Write compare_regularisation(X_train, y_train, X_val, y_val, C_values) — returns DataFrame [C, Lasso_val_acc, Ridge_val_acc, ElasticNet_val_acc, Lasso_nonzero, Ridge_nonzero]	3	PPT	--- NOT APPLICABLE ---
25	EXECUTION, Analysis and Viva	4	PPT	--- NOT APPLICABLE ---
5	implement compare_regularisation() and compare Ridge, Lasso, and Elastic Net models across different regularisation strengths while analyzing sparsity and prediction accuracy.	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 6

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching- Learning Methods	Active Learning Methods
5	Attendance and Recap	1	Talk	--- NOT APPLICABLE ---

Time(min)	Topic	BTL	Teaching- Learning Methods	Active Learning Methods
20	plot_tree visualisation; Gini vs Entropy comparison; CCP pruning path; depth sweep 1–None; dt_report() function	3	PPT	--- NOT APPLICABLE ---
25	EXECUTION, Analysis and Viva	4	PPT	--- NOT APPLICABLE ---
20	Write dt_report(X_train, y_train, X_val, y_val) — sweeps max_depth 1–15, returns DataFrame [depth, train_acc, val_acc, n_leaves], plots curves, prints best row	3	PPT	--- NOT APPLICABLE ---
25	EXECUTION, Analysis and Viva	4	PPT	--- NOT APPLICABLE ---
5	Conclusion and Summary	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 7

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching- Learning Methods	Active Learning Methods
5	Attendance and Recap	1	Talk	--- NOT APPLICABLE ---
20	OOB error convergence; max_features comparison; manual bootstrap bagging; permutation importance; rf_vs_single_tree() function	3	PPT	--- NOT APPLICABLE ---
25	Execution, Analysis and Viva	4	PPT	--- NOT APPLICABLE ---
20	Write rf_vs_single_tree(X_train, y_train, X_val, y_val, n_estimators_list) — returns DataFrame [n_estimators, RF_val_acc, DT_val_acc, OOB_acc], plots 3 lines	3	PPT	--- NOT APPLICABLE ---
25	Execution, Analysis and Viva	4	PPT	--- NOT APPLICABLE ---
5	Conclusion and Summary	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 8

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching- Learning Methods	Active Learning Methods
5	Attendance and Recap	1	Talk	--- NOT APPLICABLE ---
20	AdaBoost weights; XGBoost early stopping; LightGBM timing; SHAP beeswarm; v1 vs v2 bar chart; boosting_benchmark() function	3	PPT	--- NOT APPLICABLE ---
25	Execution, Analysis and Viva	4	PPT	--- NOT APPLICABLE ---
20	Write boosting_benchmark(X_train, y_train, X_val, y_val) — fits AdaBoost, GBC, XGB, LGB; returns DataFrame sorted by val_accuracy descending	3	PPT	--- NOT APPLICABLE ---
25	Execution, Analysis and Viva	4	PPT	--- NOT APPLICABLE ---
5	Conclusion and Summary	1	PPT	--- NOT APPLICABLE ---

SESSION NUMBER : 9

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching- Learning Methods	Active Learning Methods
5	Attendance and Recap	1	Talk	--- NOT APPLICABLE ---
20	KMeans elbow+silhouette; MiniBatchKMeans timing; hierarchical dendrograms; DBSCAN k-distance+epsilon; cluster profile table; cluster_pipeline() function	3	PPT	--- NOT APPLICABLE ---
25	Execution, Analysis and Viva	4	PPT	--- NOT APPLICABLE ---
20	Write cluster_pipeline(X_scaled, k_range) — fits KMeans for each K, returns (labels_array, summary_DataFrame[K, inertia, silhouette]), plots curves	3	PPT	--- NOT APPLICABLE ---
25	Execution, Analysis and Viva	4	PPT	--- NOT APPLICABLE ---
5	Conclusion and Summary	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 10

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching- Learning Methods	Active Learning Methods
5	Attendance and Recap	1	Talk	--- NOT APPLICABLE ---
20	PCA scree+2D+3D; t-SNE perplexity sweep; UMAP hyperparameter grid; Isolation Forest vs IsAnomaly ground truth; dim_reduction_dashboard() PNG	3	PPT	--- NOT APPLICABLE ---

Time(min)	Topic	BTL	Teaching- Learning Methods	Active Learning Methods
25	Execution, Analysis and Viva	4	PPT	--- NOT APPLICABLE ---
20	Write dim_reduction_dashboard(X_scaled, y_labels, sample_n=1000) — PCA+t-SNE+UMAP 3-panel figure saved as dim_reduction_dashboard.png	3	PPT	--- NOT APPLICABLE ---
25	Execution, Analysis and Viva	4	PPT	--- NOT APPLICABLE ---
5	Conclusion and Summary	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 11

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching- Learning Methods	Active Learning Methods
5	Attendance and Recap	1	Talk	--- NOT APPLICABLE ---
20	Leakage demo (scaler fit before vs after split); StratifiedKFold vs KFold comparison; nested CV optimism bias; full_cv_report() function	3	PPT	--- NOT APPLICABLE ---
25	Execution, Analysis and Viva	4	PPT	--- NOT APPLICABLE ---
20	Write full_cv_report(estimator, X, y, param_grid, outer_k=5, inner_k=3) — returns dict [stratified_kfold_mean, std, nested_cv_mean, non_nested_best, optimism_bias]	3	PPT	--- NOT APPLICABLE ---
25	Execution, Analysis and Viva	4	PPT	--- NOT APPLICABLE ---
5	Conclusion and Summary	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 12

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching- Learning Methods	Active Learning Methods
5	Attendance and Recap	1	Talk	--- NOT APPLICABLE ---
20	Manual TP/FP/TN/FN derivation; ROC+PR curves for LR/RF/XGB; Platt scaling calibration; McNemar's test; model_evaluation_report() function	3	PPT	--- NOT APPLICABLE ---
25	Execution, Analysis and Viva	4	PPT	--- NOT APPLICABLE ---
20	Write full_cv_report(estimator, X, y, param_grid, outer_k=5, inner_k=3) — returns dict [stratified_kfold_mean, std, nested_cv_mean, non_nested_best, optimism_bias]	3	PPT	--- NOT APPLICABLE ---
25	Execution, Analysis and Viva	4	PPT	--- NOT APPLICABLE ---
5	Conclusion and Summary	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 13

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching- Learning Methods	Active Learning Methods
5	Attendance and Recap	1	Talk	--- NOT APPLICABLE ---
20	data_loader.py + feature_pipeline.py + trainer.py; FastAPI /predict; ONNX assert; MLflow 3 runs; batch_predict.py; end_to_end_check() function	3	PPT	--- NOT APPLICABLE ---
25	Execution, Analysis and Viva	4	PPT	--- NOT APPLICABLE ---
20	Write end_to_end_check(model_path, data_path, n_batch=20) — loads model, samples n_batch rows, asserts predictions $\in \{0,1\}$ and probabilities $\in [0,1]$, prints 'Pipeline check PASSED'	3	PPT	--- NOT APPLICABLE ---
25	Execution, Analysis and Viva	4	PPT	--- NOT APPLICABLE ---
5	Conclusion and Summary	1	Talk	--- NOT APPLICABLE ---

Skilling Course DELIVERY Plan:

Skilling session no	Topics/Experiments	CO-Mapping
1	EDA and ML Lifecycle Mapping Project	CO1
2	Preprocessing Pipeline and Cleaned Dataset	CO1
3	Feature Engineering and EDA Project	CO1

Skilling session no	Topics/Experiments	CO-Mapping
4	Linear Regression and Regularisation Project	CO2
5	Full Logistic Regression Pipeline Project	CO2
6	Multinomial Logistic Regression Project	CO2
7	Linear Regression, Feature Scaling, and Encoding Project	CO2
8	Decision Tree Classification Project	CO3
9	Random Forest Ensemble Project	CO3
10	XGBoost, LightGBM, and SHAP Interpretability Project	CO3
11	Decision Tree and Random Forest Classification Project	CO3
12	XGBoost, LightGBM, and SHAP Feature Explanation Project	CO3
13	Clustering Project	CO4
14	PCA, t-SNE, and UMAP Visualisation Project	CO4
15	Isolation Forest Anomaly Detection Project	CO4
16	Clustering and PCA Socioeconomic Profiling Project	CO4
17	DBSCAN and Isolation Forest Anomaly Detection Project	CO4
18	Cross-Validation, Metrics, and Calibration Project	CO5
19	Hyperparameter Tuning and Statistical Significance Project	CO5
20	Evaluation, Threshold Tuning, and Calibration Project	CO5
21	GridSearch, RandomSearch, Validation Curves, and Learning Curves	CO5
22	Modular Pipeline and FastAPI Deployment Project	CO6
23	MLflow, DVC, and Drift Simulation End-to-End MLOps Project	CO6
24	End-to-End MLOps: Batch Scoring, PSI Monitoring, and Retraining Pipeline	CO6

Skilling Session wise Teaching – Learning Plan

SESSION NUMBER : 1

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching- Learning Methods	Active Learning Methods
5	Attendance and Recap	1	Talk	--- NOT APPLICABLE ---
20	Load Titanic; missing heatmap; survival by sex/pclass/age; lifecycle mapping dict; correlation heatmap; titanic_eda_report.pdf export	3	PPT	--- NOT APPLICABLE ---
25	Execution, Analysis and Viva	3	PPT	--- NOT APPLICABLE ---
20	Export 4-page PDF using PdfPages: missingness heatmap, 3 Q3 plots, correlation heatmap, lifecycle mapping text. Save as titanic_eda_report.pdf	3	PPT	--- NOT APPLICABLE ---
25	Execution, Analysis and Viva	4	PPT	--- NOT APPLICABLE ---
5	Conclusion and Summary	1	PPT	--- NOT APPLICABLE ---

SESSION NUMBER : 2

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching- Learning Methods	Active Learning Methods
5	Attendance and Recap	1	Talk	--- NOT APPLICABLE ---
20	Drop leakage columns; impute age+embarked; OHE sex+embarked; OrdinalEncoder pclass; is_alone feature; Pipeline validation function	2	PPT	--- NOT APPLICABLE ---

Time(min)	Topic	BTL	Teaching- Learning Methods	Active Learning Methods
25	Evaluation, Analysis & viva	3	PPT	--- NOT APPLICABLE ---
20	Write build_Project_pipeline() and validate_pipeline(pipeline, X_train, X_val) — asserts no NaN, correct shapes, >5 columns; prints 'Pipeline validation PASSED'	3	PPT	--- NOT APPLICABLE ---
25	Evaluation, Analysis & viva	4	PPT	--- NOT APPLICABLE ---
5	Conclusion and Summary	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 3

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching- Learning Methods	Active Learning Methods
5	Attendance and Recap	1	Talk	--- NOT APPLICABLE ---
20	fetch_Project; '?' → NaN; income by education/occupation/hours-per-week; Pearson correlation; adult_eda_report() top-3 features	3	PPT	--- NOT APPLICABLE ---
25	Execution, Analysis and Viva	4	PPT	--- NOT APPLICABLE ---
20	Write adult_eda_report(df) — prints missing summary, class distribution, mean hours-per-week per income group; returns top-3 features by absolute correlation with target	3	PPT	--- NOT APPLICABLE ---
25	Execution, Analysis and Viva	4	PPT	--- NOT APPLICABLE ---
5	Conclusion and Summary	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 4

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching- Learning Methods	Active Learning Methods
5	Attendance and Recap	1	Talk	--- NOT APPLICABLE ---
20	Fetch Project Dataset; NumPy GD vs sklearn; Ridge/Lasso RMSE curves; ElasticNet comparison table; coefficient bar charts; regression_benchmark()	3	PPT	--- NOT APPLICABLE ---
25	Execution, Analysis and Viva	4	PPT	--- NOT APPLICABLE ---
20	Write regression_benchmark(X_train, y_train, X_test, y_test, alphas) — fits LR/Ridge/Lasso/ElasticNet; returns DataFrame sorted by test RMSE ascending	3	PPT	--- NOT APPLICABLE ---
25	Execution, Analysis and Viva	4	PPT	--- NOT APPLICABLE ---
5	Conclusion and Summary	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 5

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching- Learning Methods	Active Learning Methods
5	Attendance and Recap	1	Talk	--- NOT APPLICABLE ---
20	70/15/15 split; LR with/without scaling (n_iter_comparison); C tuning; test accuracy; coefficient chart; logistic_pipeline_report()	3	PPT	--- NOT APPLICABLE ---
25	Execution, Analysis and Viva	4	PPT	--- NOT APPLICABLE ---
20	Write logistic_pipeline_report(X_train, y_train, X_val, y_val, C_values) — tunes C via val accuracy; prints train/val/test accuracy; returns sorted coefficient Series	3	PPT	--- NOT APPLICABLE ---
25	Execution, Analysis and Viva	4	PPT	--- NOT APPLICABLE ---
5	Conclusion and Summary	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 6

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching- Learning Methods	Active Learning Methods
5	Attendance and Recap	1	Talk	--- NOT APPLICABLE ---
20	Zero→NaN replacement; Glucose binning (Low/Mid/High); multinomial LR; per-class accuracy vs C; coefficient subplots; multinomial_analysis()	3	PPT	--- NOT APPLICABLE ---
25	Execution, Analysis and Viva	4	PPT	--- NOT APPLICABLE ---

Time(min)	Topic	BTL	Teaching- Learning Methods	Active Learning Methods
20	Write multinomial_analysis(X_train, y_train, X_val, y_val, class_names, C_values) — records per-class F1 per C; returns DataFrame [C, overall_acc, F1_Low, F1_Mid, F1_High]	3	PPT	--- NOT APPLICABLE ---
25	Execution, Analysis and Viva	4	PPT	--- NOT APPLICABLE ---
5	Conclusion and Summary	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 7

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching- Learning Methods	Active Learning Methods
5	Attendance and Recap	1	Talk	--- NOT APPLICABLE ---
20	Execution, Analysis and Viva	3	PPT	--- NOT APPLICABLE ---
25	Actual vs predicted scatter; OrdinalEncoder origin; scaling impact on RMSE; power_to_weight + age features; mpg_regression_pipeline()	4	PPT	--- NOT APPLICABLE ---
20	Execution, Analysis and Viva	3	PPT	--- NOT APPLICABLE ---
25	Write Project_regression_pipeline(df, target_col='mpg') — drops missing, encodes, scales, splits 80/20; fits LR/Ridge/Lasso; returns DataFrame [model, train_rmse, test_rmse, test_r2]	4	PPT	--- NOT APPLICABLE ---
5	Conclusion and Summary	1	PPT	--- NOT APPLICABLE ---

SESSION NUMBER : 8

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching- Learning Methods	Active Learning Methods
5	Attendance and Recap	1	Talk	--- NOT APPLICABLE ---
20	Full tree overfitting demo; Gini vs Entropy; DT Regressor on Age (RMSE); CCP pruning; depth sweep annotated; dt_full_analysis()	3	PPT	--- NOT APPLICABLE ---
25	Execution, Analysis and Viva	4	PPT	--- NOT APPLICABLE ---
20	Write dt_full_analysis(X_train, y_train, X_val, y_val) — returns dict [full_tree_val_acc, optimal_depth_val_acc, pruned_val_acc, best_depth, optimal_alpha]	3	PPT	--- NOT APPLICABLE ---
25	Execution, Analysis and Viva	4	PPT	--- NOT APPLICABLE ---
5	Conclusion and Summary	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 9

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching- Learning Methods	Active Learning Methods
5	Attendance and Recap	1	Talk	--- NOT APPLICABLE ---
20	OOB convergence; manual bagging 5 trees; max_features bar chart; permutation vs Gini ranking; ensemble_comparison() DataFrame	3	PPT	--- NOT APPLICABLE ---
25	Execution, Analysis and Viva	4	PPT	--- NOT APPLICABLE ---
20	Write ensemble_comparison(X_train, y_train, X_val, y_val) — fits DT (best depth), manual bagged (10 trees), RF(200); returns DataFrame sorted by val_accuracy	3	PPT	--- NOT APPLICABLE ---
25	Execution, Analysis and Viva	4	PPT	--- NOT APPLICABLE ---
5	Conclusion and Summary	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 10

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching- Learning Methods	Active Learning Methods
5	Attendance and Recap	1	Talk	--- NOT APPLICABLE ---
20	AdaBoost weights; GBC loss curve; XGBoost early stopping; 6-row model comparison table; SHAP beeswarm+waterfall; shap_feature_summary()	3	PPT	--- NOT APPLICABLE ---
25	Execution, Analysis and Viva	4	PPT	--- NOT APPLICABLE ---
20	Write shap_feature_summary(model, X_val, feature_names, top_n=5) — returns DataFrame [feature, mean_abs_shap] for top_n features sorted descending	3	PPT	--- NOT APPLICABLE ---

Time(min)	Topic	BTL	Teaching- Learning Methods	Active Learning Methods
25	Execution, Analysis and Viva	4	PPT	--- NOT APPLICABLE ---
5	Conclusion and Summary	3	PPT	--- NOT APPLICABLE ---

SESSION NUMBER : 11

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching- Learning Methods	Active Learning Methods
5	Attendance and Recap	1	Talk	--- NOT APPLICABLE ---
20	Quality binarisation (≥ 6); full-tree overfitting; CCP pruning; OOB convergence; permutation importance; wine_model_comparison()	3	PPT	--- NOT APPLICABLE ---
25	Execution, Analysis and Viva	4	PPT	--- NOT APPLICABLE ---
20	Write Project_model_comparison(X_train, y_train, X_val, y_val) — fits DT (optimal depth), RF (n_estimators=200), RF (top-5 permutation features); returns DataFrame [model, val_accuracy, n_features_used]	3	PPT	--- NOT APPLICABLE ---
25	Execution, Analysis and Viva	4	PPT	--- NOT APPLICABLE ---
5	Conclusion and Summary	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 12

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching- Learning Methods	Active Learning Methods
5	Attendance and Recap	1	Talk	--- NOT APPLICABLE ---
20	OHE categorical cols; AdaBoost+GBC; XGBoost early stopping; Gini vs permutation importance; SHAP beeswarm; boosting_pipeline_report()	3	PPT	--- NOT APPLICABLE ---
25	Execution, Analysis and Viva	4	PPT	--- NOT APPLICABLE ---
20	Write boosting_pipeline_report(X_train, y_train, X_val, y_val) — fits 4 boosting models; returns tuple (model_df[model, val_accuracy, time_s], top5_shap_df[feature, mean_abs_shap])	3	PPT	--- NOT APPLICABLE ---
25	Execution, Analysis and Viva	4	PPT	--- NOT APPLICABLE ---
5	Conclusion and Summary	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 13

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching- Learning Methods	Active Learning Methods
5	Attendance and Recap	1	Talk	--- NOT APPLICABLE ---
20	KMeans elbow+silhouette; inverse-transform centroids scatter; DBSCAN k-distance; Ward dendrogram; ARI score; customer_segment_report()	3	PPT	--- NOT APPLICABLE ---
25	Execution, Analysis and Viva	4	PPT	--- NOT APPLICABLE ---
20	Write Project_report(df, feature_cols, k) — scales features, fits KMeans(k), computes per-cluster means; returns styled DataFrame with highest value highlighted	3	PPT	--- NOT APPLICABLE ---
25	Execution, Analysis and Viva	4	PPT	--- NOT APPLICABLE ---
5	Conclusion and Summary	1	PPT	--- NOT APPLICABLE ---

SESSION NUMBER : 14

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching- Learning Methods	Active Learning Methods
5	Attendance and Recap	1	Talk	--- NOT APPLICABLE ---
20	PCA scree; t-SNE 4-perplexity grid; UMAP KMeans+Spending coloured; 3-panel PNG; compare_dr_methods() silhouette dict	3	PPT	--- NOT APPLICABLE ---
25	Execution, Analysis and Viva	4	PPT	--- NOT APPLICABLE ---
20	Write compare_dr_methods(X_scaled, labels, perplexity=30, n_neighbors=10) — returns dict {'PCA_silhouette':..., 'TSNE_silhouette':..., 'UMAP_silhouette':...}	3	PPT	--- NOT APPLICABLE ---
25	Execution, Analysis and Viva	4	PPT	--- NOT APPLICABLE ---

Time(min)	Topic	BTL	Teaching- Learning Methods	Active Learning Methods
5	Conclusion and Summary	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 15

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching- Learning Methods	Active Learning Methods
5	Attendance and Recap	1	Talk	--- NOT APPLICABLE ---
20	10k stratified sample; IsolationForest precision/recall; contamination sweep F1; score histogram; OneClassSVM comparison; anomaly_benchmark()	3	PPT	--- NOT APPLICABLE ---
25	Execution, Analysis and Viva	4	PPT	--- NOT APPLICABLE ---
20	Write anomaly_benchmark(X, y_true, contamination_values) — fits IsolationForest per contamination; returns DataFrame [contamination, precision, recall, F1] sorted by F1 desc	3	PPT	--- NOT APPLICABLE ---
25	Execution, Analysis and Viva	4	PPT	--- NOT APPLICABLE ---
5	Conclusion and Summary	1	PPT	--- NOT APPLICABLE ---

SESSION NUMBER : 16

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching- Learning Methods	Active Learning Methods
5	Attendance and Recap	1	Talk	--- NOT APPLICABLE ---
20	write Project_clustering; KMeans cluster personas; dendrogram; ARI; PCA 2D annotated ; Project_profile()	3	PPT	--- NOT APPLICABLE ---
25	Execution, Analysis and Viva	4	PPT	--- NOT APPLICABLE ---
20	Write Project_profile(df, feature_cols, k, region_col=None) — scales, fits KMeans(k), computes per-cluster means + most frequent region; returns styled DataFrame	3	PPT	--- NOT APPLICABLE ---
25	Execution, Analysis and Viva	4	PPT	--- NOT APPLICABLE ---
5	Conclusion and Summary	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 17

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching- Learning Methods	Active Learning Methods
5	Attendance and Recap	1	Talk	--- NOT APPLICABLE ---
20	log1p transform; k-distance epsilon; DBSCAN (eps,min_samples) sweep table; IsolationForest score histogram; intrusion_detection_report()	3	PPT	--- NOT APPLICABLE ---
25	Execution, Analysis and Viva	4	PPT	--- NOT APPLICABLE ---
20	Write intrusion_detection_report(X_scaled, y_true, eps_values, contamination_values) — DBSCAN+IsolationForest F1 per parameter; returns DataFrame sorted by F1 descending	3	PPT	--- NOT APPLICABLE ---
25	Execution, Analysis and Viva	4	PPT	--- NOT APPLICABLE ---
5	Conclusion and Summary	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 18

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching- Learning Methods	Active Learning Methods
5	Attendance and Recap	1	Talk	--- NOT APPLICABLE ---
20	10-fold CV (accuracy/F1/ROC-AUC); ROC+PR curves; learning curve shaded; isotonic calibration; Brier score; calibration_comparison()	3	PPT	--- NOT APPLICABLE ---
25	Execution, Analysis and Viva	4	PPT	--- NOT APPLICABLE ---
20	Write calibration_comparison(models_dict, X_train, y_train, X_val, y_val) — Brier before/after isotonic calibration; returns DataFrame [model, brier_before, brier_after, improvement]	3	PPT	--- NOT APPLICABLE ---
25	Execution, Analysis and Viva	4	PPT	--- NOT APPLICABLE ---
5	Conclusion and Summary	1	PPT	--- NOT APPLICABLE ---

SESSION NUMBER : 19

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching- Learning Methods	Active Learning Methods
5	Attendance and Recap	1	Talk	--- NOT APPLICABLE ---
20	GridSearchCV; RandomizedSearchCV histogram; BayesSearchCV comparison table; validation curves; McNemar's test; tuning_summary()	3	PPT	--- NOT APPLICABLE ---
25	Execution, Analysis and Viva	4	PPT	--- NOT APPLICABLE ---
20	Write tuning_summary(X_train, y_train, X_val, y_val) — runs Grid/Random/BayesSearch (n_iter=20 each); returns DataFrame [method, best_roc_auc, n_fits, time_s]	3	PPT	--- NOT APPLICABLE ---
25	Execution, Analysis and Viva	4	PPT	--- NOT APPLICABLE ---
5	Conclusion and Summary	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 20

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching- Learning Methods	Active Learning Methods
5	Attendance and Recap	1	Talk	--- NOT APPLICABLE ---
20	load_Project(); grouped metric bar chart; recall $\geq$ 0.95 threshold; ROC+PR curves; Platt scaling reliability; threshold_analysis()	3	PPT	--- NOT APPLICABLE ---
25	Execution, Analysis and Viva	4	PPT	--- NOT APPLICABLE ---
20	Write threshold_analysis(model, X_val, y_val, recall_target=0.95) — sweeps threshold 0.05–0.95; returns DataFrame with all rows; prints lowest threshold achieving recall $\geq$ recall_target	3	PPT	--- NOT APPLICABLE ---
25	Execution, Analysis and Viva	4	PPT	--- NOT APPLICABLE ---
5	Conclusion and Summary	1	PPT	--- NOT APPLICABLE ---

SESSION NUMBER : 21

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching- Learning Methods	Active Learning Methods
5	Attendance and Recap	1	Talk	--- NOT APPLICABLE ---
20	fetch_openml banknote-authentication; GridSearchCV; RandomSearch histogram; n_estimators+max_depth validation curves; learning curve; search_comparison()	3	PPT	--- NOT APPLICABLE ---
25	Execution, Analysis and Viva	4	PPT	--- NOT APPLICABLE ---
20	Write search_comparison(estimator, param_grid, X_train, y_train, n_iter=30) — runs GridSearchCV + RandomizedSearchCV; returns DataFrame [method, best_score, n_fits, time_s]	3	PPT	--- NOT APPLICABLE ---
25	Execution, Analysis and Viva	4	PPT	--- NOT APPLICABLE ---
5	Conclusion and Summary	1	PPT	--- NOT APPLICABLE ---

SESSION NUMBER : 22

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching- Learning Methods	Active Learning Methods
5	Attendance and Recap	1	Talk	--- NOT APPLICABLE ---
20	LR/RF/DT val accuracy; Project_model.joblib; FastAPI /health + /predict; pytest 5 tests; batch_Project.py; api_health_check()	3	PPT	--- NOT APPLICABLE ---
25	Execution, Analysis and Viva	4	PPT	--- NOT APPLICABLE ---
20	Write api_health_check(base_url, test_payloads) — calls POST /predict for each payload, asserts status=200 and keys present; returns results DataFrame	3	PPT	--- NOT APPLICABLE ---
25	Execution, Analysis and Viva	4	PPT	--- NOT APPLICABLE ---
5	Conclusion and Summary	1	PPT	--- NOT APPLICABLE ---

SESSION NUMBER : 23

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching- Learning Methods	Active Learning Methods
5	Attendance and Recap	1	Talk	--- NOT APPLICABLE ---
20	3 MLflow runs; DVC version rollback; PSI per feature; drift dashboard PNG; retrain_if_drifted(); psi_report() DataFrame	3	PPT	--- NOT APPLICABLE ---
25	Execution, Analysis and Viva	4	PPT	--- NOT APPLICABLE ---
20	Write psi_report(reference_df, production_df, num_cols, threshold=0.2) — returns DataFrame [feature, PSI, status='DRIFTED'/'STABLE']; prints summary (n_drifted, n_stable)	3	PPT	--- NOT APPLICABLE ---
25	Execution, Analysis and Viva	4	PPT	--- NOT APPLICABLE ---
5	Conclusion and Summary	1	PPT	--- NOT APPLICABLE ---

SESSION NUMBER : 24

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching- Learning Methods	Active Learning Methods
5	Attendance and Recap	1	Talk	--- NOT APPLICABLE ---
20	wine_model.joblib; batch_wine.py 50-row CSV; PSI alcohol+volatile acidity drift; monitoring dashboard PNG; 3 MLflow runs; wine_retrain_pipeline()	3	PPT	--- NOT APPLICABLE ---
25	Execution, Analysis and Viva	4	PPT	--- NOT APPLICABLE ---
20	Write Project_retrain_pipeline(reference_csv, production_csv, model_path, psi_threshold=0.2) — computes PSI; if drifted retrains and logs to MLflow; returns dict [drifted_features, retrained, new_model_path]	3	PPT	--- NOT APPLICABLE ---
25	Execution, Analysis and Viva	4	PPT	--- NOT APPLICABLE ---
5	Conclusion and Summary	1	PPT	--- NOT APPLICABLE ---

WEEKLY HOMEWORK ASSIGNMENTS/ PROBLEM SETS/OPEN ENDED PROBLEM-SOLVING EXERCISES etc:

Week	Assignment Type	Assignment No	Topic	Details	co
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COURSE TIME TABLE:

	Hour	1	2	3	4	5	6	7	8	9
Day	Component									
Mon	Theory	H-S7,H-S8,H-S9,H-S14,H-S15	H-S7,H-S8,H-S9,H-S14,H-S15	---	---	---	-- -	---	--	--
	Tutorial	--	--	---	---	---	-- -	---	--	--
	Lab	H-S12,H-S12,H-S13,H-S13	H-S12,H-S12,H-S13,H-S13	---	---	---	-- -	---	--	--
	Skilling	--	--	---	---	---	-- -	---	H-S7,H-S7,H-S8,H-S8,H-S9,H-S9,H-S10,H-S10,H-S11,H-S11	H-S7,H-S7,H-S8,H-S8,H-S9,H-S9,H-S10,H-S10,H-S11,H-S11
Tue	Theory	H-S1,H-S2,H-S3	H-S1,H-S2,H-S3	--	--	--	-- -	--	--	--
	Tutorial	--	--	--	--	--	-- -	--	--	--
	Lab	--	--	--	--	H-S4,H-S4,H-S5,H-S5,H-S6,H-S6,H-S14,H-S14,H-S15,H-S15	H-S4,H-S4,H-S5,H-S5,H-S6,H-S6,H-S14,H-S14,H-S15,H-S15	--	--	--
	Skilling	--	--	H-S12,H-S12,H-S13,H-S13	H-S12,H-S12,H-S13,H-S13	--	-- -	--	H-S1,H-S1,H-S2,H-S2,H-S3,H-S3	H-S1,H-S1,H-S2,H-S2,H-S3,H-S3
Wed	Theory	---	---	H-S4,H-S5,H-S6	H-S4,H-S5,H-S6	--	-- -	--	---	---
	Tutorial	---	---	--	--	--	-- -	--	---	---
	Lab	---	---	--	--	--	-- -	--	---	---
	Skilling	---	---	H-S14,H-S14,H-S15,H-S15	H-S14,H-S14,H-S15,H-S15	H-S10,H-S10,H-S11,H-S11	-- -	H-S10,H-S10,H-S11,H-S11	---	---

	Hour	1	2	3	4	5	6	7	8	9
Thu	Theory	---	---	--	--	--	--	---	---	---
	Tutorial	---	---	--	--	--	--	---	---	---
	Lab	---	---	H-S10,H-S10,H-S11,H-S11	H-S10,H-S10,H-S11,H-S11	--	--	---	---	---
	Skilling	---	---	--	--	H-S4,H-S4,H-S5,H-S5,H-S6,H-S6	--	H-S4,H-S4,H-S5,H-S5,H-S6,H-S6	---	---
Fri	Theory	H-S10,H-S11	H-S10,H-S11	--	H-S12,H-S13	H-S12,H-S13	--	--	--	--
	Tutorial	--	--	--	--	--	--	--	--	--
	Lab	--	--	--	--	--	--	--	H-S1,H-S1,H-S2,H-S2,H-S3,H-S3	H-S1,H-S1,H-S2,H-S2,H-S3,H-S3
	Skilling	--	--	H-S4,H-S4,H-S5,H-S5,H-S6,H-S6	H-S4,H-S4,H-S5,H-S5,H-S6,H-S6	H-S7,H-S7,H-S8,H-S8,H-S9,H-S9	--	H-S7,H-S7,H-S8,H-S8,H-S9,H-S9	--	--
Sat	Theory	---	---	--	--	--	--	--	--	--
	Tutorial	---	---	--	--	--	--	--	--	--
	Lab	---	---	H-S7,H-S7,H-S8,H-S8,H-S9,H-S9	H-S7,H-S7,H-S8,H-S8,H-S9,H-S9	--	--	--	--	--
	Skilling	---	---	--	H-S12,H-S12,H-S13,H-S13	H-S1,H-S1,H-S2,H-S2,H-S3,H-S3,H-S12,H-S12,H-S13,H-S13	--	H-S1,H-S1,H-S2,H-S2,H-S3,H-S3	H-S14,H-S14,H-S15,H-S15	H-S14,H-S14,H-S15,H-S15
Sun	Theory	--	--	--	--	--	--	--	--	--
	Tutorial	--	--	--	--	--	--	--	--	--
	Lab	--	--	--	--	--	--	--	--	--
	Skilling	--	--	--	--	--	--	--	--	--

REMEDIAL CLASSES:

Supplementary course handouts, including special lectures and discussions, will be planned and the schedule will be notified accordingly.

SELF-LEARNING:

Assignments to promote self-learning, survey of contents from multiple sources.

S.no	Topics	CO	ALM	References/MOOCs
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DELIVERY DETAILS OF CONTENT BEYOND SYLLABUS:

Content beyond syllabus covered (if any) should be delivered to all students that would be planned, and schedule notified accordingly.

S.no	Advanced Topics, Additional Reading, Research papers and any	CO	ALM	References/MOOCs
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EVALUATION PLAN:

Evaluation Type	Evaluation Component	Weightage/Marks		Assessment Dates	Duration (Minutes)	CO1	CO2	CO3	CO4	CO5	CO6
End Semester Summative Evaluation Total= 40 %	Term End Project Defence & Demonstration	Weightage	16		180	3	2.5	2.5	2.5	2.5	3
		Max Marks	60			10	10	10	10	10	10
	End Semester Exam	Weightage	24		180	2	2	2	6	6	6
		Max Marks	90			10	10	10	20	20	20
In Semester Formative Evaluation Total= 36 %	MOOCs Progress Review	Weightage	8		150	1.5	1.5	1.5	1.5	1	1
		Max Marks	150			25	25	25	25	25	25
	Global Challenges - Participation	Weightage	8		150	1	1	1.5	1.5	1.5	1.5
		Max Marks	150			25	25	25	25	25	25
	Project Continuous Evaluation	Weightage	8		150	1.5	1.5	1	1	1.5	1.5
		Max Marks	150			25	25	25	25	25	25
	Hackathon	Weightage	12		150	2	2	2	2	2	2
		Max Marks	150			25	25	25	25	25	25
In Semester Summative	In-Term Skill-based Examination	Weightage	12		120	2	2	2	2	2	2
		Max Marks	60			10	10	10	10	10	10

Evaluation Type	Evaluation Component	Weightage/Marks		Assessment Dates	Duration (Minutes)	CO1	CO2	CO3	CO4	CO5	CO6
Evaluation Total= 24 %	Mid-Term Examination (Descriptive)	Weightage	12		90	4	4	4			
		Max Marks	45			15	15	15			

ATTENDANCE POLICY:

Every student is expected to be responsible for regularity of his/her attendance in class rooms and laboratories, to appear in scheduled tests and examinations and fulfill all other tasks assigned to him/her in every course

In every course, student has to maintain a minimum of 85% attendance to be eligible for appearing in Semester end examination of the course, for cases of medical issues and other unavoidable circumstances the students will be condoned if their attendance is between 75% to 85% in every course, subjected to submission of medical certificates, medical case file and other needful documental proof to the concerned departments

DETENTION POLICY :

In any course, a student has to maintain a minimum of 85% attendance and In-Semester Examinations to be eligible for appearing to the Semester End Examination, failing to fulfill these conditions will deem such student to have been detained in that course.

PLAGIARISM POLICY :

Plagiarism is treated as a serious academic offense, and the university enforces a strict zero-tolerance policy to uphold academic integrity. For a first offense, minor cases may result in a warning and reduced marks, while major cases may lead to a zero grade and mandatory academic integrity training. A second offense will result in a zero grade, formal reprimand, and academic probation, whereas a third offense may lead to suspension and a permanent record of misconduct. Repeated violations can result in expulsion, and in group work, all members are held accountable unless responsibility is clearly established.

COURSE TEAM MEMBERS, CHAMBER CONSULTATION HOURS AND CHAMBER VENUE DETAILS:

Name of Faculty	Delivery Component of Faculty	Sections of Faculty	Chamber Consultation Day (s)	Chamber Consultation Timings for each day	Chamber Consultation Room No:	Signature of Course faculty:
Rani Medidha	L	5-MA,9-MA	-	-	-	-
Rani Medidha	P	9-A,5-A,1-B	-	-	-	-
Rani Medidha	S	3-B,5-A,9-A	-	-	-	-
Dyuti Banerjee	S	13-B	-	-	-	-
Gunturi Raghavendra	L	1-MA	-	-	-	-
Gunturi Raghavendra	P	9-B,6-B,1-A	-	-	-	-
Gunturi Raghavendra	S	1-A,6-B,9-B	-	-	-	-
Venu Vinnakota	L	15-MA	-	-	-	-
Venu Vinnakota	P	15-A	-	-	-	-
Venu Vinnakota	S	15-A	-	-	-	-
Nelapati Lava Prasad	L	14-MA	-	-	-	-
Nelapati Lava Prasad	P	14-A	-	-	-	-
Nelapati Lava Prasad	S	14-A	-	-	-	-
GADDAM RAVINDRA BABU	L	6-MA,7-MA	-	-	-	-
GADDAM RAVINDRA BABU	P	7-A,6-A	-	-	-	-
GADDAM RAVINDRA BABU	S	7-A,6-A,1-B	-	-	-	-
Surendra Kota	P	14-B	-	-	-	-
Surendra Kota	S	14-B	-	-	-	-
ASIF SHAIK	L	8-MA,3-MA	-	-	-	-
ASIF SHAIK	P	8-A,3-A	-	-	-	-
ASIF SHAIK	S	8-A,5-B,3-A	-	-	-	-
khaja sk	S	12-B	-	-	-	-
Venkateswara Rao Pulipati	P	4-B,3-B	-	-	-	-
Anitha Chekuri	P	10-B,13-B	-	-	-	-
Anitha Chekuri	S	13-B	-	-	-	-
Nirmalajyothi Narisetty	L	2-MA	-	-	-	-
Nirmalajyothi Narisetty	P	2-A,7-B	-	-	-	-
Nirmalajyothi Narisetty	S	2-A,4-B,7-B	-	-	-	-
LakshmiPrasanna Yeluri	L	10-MA	-	-	-	-
LakshmiPrasanna Yeluri	P	10-A	-	-	-	-
LakshmiPrasanna Yeluri	S	10-A	-	-	-	-
Chandusha kanda	S	12-B	-	-	-	-
SRAVYA RAJAMANURI VENKATA	P	15-B	-	-	-	-
SRAVYA RAJAMANURI VENKATA	S	15-B	-	-	-	-
Jyothsna Datti	L	4-MA	-	-	-	-
Jyothsna Datti	P	4-A,2-B,8-B	-	-	-	-
Jyothsna Datti	S	8-B,4-A,2-B	-	-	-	-

Name of Faculty	Delivery Component of Faculty	Sections of Faculty	Chamber Consultation Day (s)	Chamber Consultation Timings for each day	Chamber Consultation Room No:	Signature of Course faculty:
I V SAI HARITHA	L	11-MA,13-MA	-	-	-	-
I V SAI HARITHA	P	11-A,13-A	-	-	-	-
I V SAI HARITHA	S	11-A,13-A	-	-	-	-
pavani Chilumula	P	5-B	-	-	-	-
pavani Chilumula	S	3-B	-	-	-	-
Raghupathi Manthena	S	10-B	-	-	-	-
Soumya Erukonda	P	12-B,11-B	-	-	-	-
Soumya Erukonda	S	11-B	-	-	-	-
Divya Priya Degala	S	12-B,10-B	-	-	-	-
MEKALA SANJEEV KUMAR	L	12-MA	-	-	-	-
MEKALA SANJEEV KUMAR	P	12-A	-	-	-	-
MEKALA SANJEEV KUMAR	S	12-A	-	-	-	-

GENERAL INSTRUCTIONS

Students should come prepared for classes by reviewing the prescribed video lectures in advance and carrying the required textbook(s) or study material(s) as specified by the Course Faculty.

NOTICES

All course-related notices and communications will be shared through the official institutional email.

Signature of COURSE COORDINATOR

(LakshmiPrasanna Yeluri)(9041)(CSE-1)

Signature of Department Prof. Incharge Academics & Vetting Team Member

Department Of CSE-1

HEAD OF DEPARTMENT:**Approval from: DEAN-ACADEMICS**

(Sign with Office Seal)

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